2016.0 RANGE ROVER (LG), 414-00

BATTERY AND CHARGING SYSTEM – GENERAL INFORMATION

DESCRIPTION AND OPERATION

1. INTRODUCTION

This document defines the requirements for care and maintenance of batteries, and the standard of battery care at retailers for new vehicles.

This applies to all types of 12 Volt Lead Acid Batteries used in Jaguar and Land Rover vehicles whether they are conventional flooded technology or Absorbed Glass Mat (AGM – also known as Valve Regulated Lead Acid (VRLA)) technology and also applies to both Primary, Secondary and Auxiliary Batteries. AGM batteries offer improved resistance to cycling as seen in stop start applications.

In order to prevent damage to the battery and ensure a satisfactory service life, all processes detailed within this document must be rigorously adhered to.

It is equally important therefore to note the following key points:

- All new vehicles leave the factory with either a transit relay installed and/or have a transit mode programmed into the vehicle control modules. The transit relay must be removed and the transit mode disabled (where applicable) using an approved diagnostic system, **NOT MORE THAN 72**

HOURS before the customer takes delivery.

- The battery can be discharged by the following mechanisms:
  - **Self Discharge**: - A lead acid battery will very slowly discharge itself due to its own internal chemical processes whether it is connected to a vehicle or not.
  - **Quiescent Discharge**: - The vehicle electrical systems when connected to the battery will draw charge from the battery.

12 Volt Lead Acid Batteries rely on internal chemical processes to create a voltage and deliver current. These processes and the internal chemical structure of the battery can be damaged if the battery is allowed to discharge over a number of weeks / months, or is left in a discharged state for a lengthy time period.

- **On vehicles with conventional ignition keys**, these must not be left in the ignition lock barrel when the transit relay has been removed, otherwise quiescent current will increase and the battery will discharge more rapidly.
- **For keyless vehicles**, the Smart Key must be stored at least 5m (16 ft) away from the vehicle when the vehicle is parked or stored.
- **AGM Batteries are fully sealed and cannot have the electrolyte level topped up.**

NOTE:

> Retailers involved in the storage / handling of vehicles and replacement batteries have a responsibility to ensure that only a fully charged battery may be processed through the distribution selling chain.

2. GENERAL RULES FOR BATTERY CARE

2.1 RETAILER SHOWROOM/DEMONSTRATION VEHICLES

Vehicles used as retailer demonstrator(s), in a showroom, must be connected to a JLR approved showroom conditioner capable of delivering 50 Amps. This will prevent the battery from being damaged.

Vehicles used as retailer demonstrator(s), in a showroom, must be taken out of transit mode or have the transit relay removed. For additional information on the transit relay installation, refer to the **Transit Relay Installation Summary table**.

Vehicles used as retailer demonstrator(s), in a showroom, must have the starter motor, horn and wiper fuses removed as a minimum.
For additional information, refer to: Showroom Preparation (101-02 Showroom Preparation, Description and Operation).

WARNING:

> Showroom conditioners are only required to be turned on during showroom opening hours, outside these hours switch off all showroom conditioners.

NOTE:

> For retailers who do not have floor sockets installed in the showroom a slave battery process has been outlined below. The slave battery process is only a TEMPORARY process until floor sockets are installed into the showroom. The retailer is responsible for making sure the process is set up and is also responsible for the procurement of slave batteries and other supporting equipment. The retailer is also responsible for developing a conformance plan to install floor sockets.

2.1.1 RECOMMENDED SLAVE BATTERY RESPONSIBILITIES/PROCESS

Showroom Representative Responsibilities

1. All vehicles in the retailer showroom that are being used for display purposes **MUST** be fitted with a slave battery if the JLR approved battery conditioner cannot be installed.
2. All vehicles fitted with a slave battery must be logged and tracked by a nominated showroom representative who has been trained in the Battery Care Requirements process. For additional information, Refer to the following Excellence training courses Battery Care Requirements EDASS00025_0814 and Good Battery Care EDAS400059_1114
3. A list of the vehicle VIN details must be provided to the department where the original vehicle battery will be stored and maintained
4. Make sure the installation of the slave battery is coordinated with the workshop department and installed in a timely manner
5. Check on a daily basis for a low battery warning display on the vehicle information display. If required, contact the workshop department to charge the slave battery.
6. Make sure that the slave battery condition is checked on a weekly basis using the Midtronics EXP-1080. If required, contact the workshop department to charge the slave battery
7. When the display vehicle has been sold, notify the workshop department to install the original battery (in a fully charged state) within 72 hours of vehicle handover to the customer
8. When the display vehicle is being returned to the vehicle storage compound, notify the workshop department to install the original battery (in a fully charged state), refit the transit relay (if required) and put the vehicle into transportation mode
9. Record all battery inspections and battery charging dates and supporting information on the showroom and slave battery report form.
   For additional information, refer to: Showroom and Slave Battery

Condition Report Form (414-00 Battery and Charging System - General Information, Description and Operation).

WORKSHOP REPRESENTATIVE RESPONSIBILITIES

1. Removal of the original vehicle battery from the vehicle
2. Installation of the slave battery to the display vehicle
3. Clearly label and identify a slave battery
4. Attach the slave battery identification tag JLR-415-010 (Available from the Jaguar Landrover equipment website) onto the vehicle keys identifying that the vehicle is fitted with a slave battery
5. Check, maintain and record the original battery condition information in accordance with the JLR storage procedure and policy.
   For additional information, refer to: Showroom and Slave Battery Condition Report Form (414-00 Battery and Charging System - General Information, Description and Operation).
6. Make sure the slave battery has the same part number as the original battery fitted to the display vehicle.
7. Make sure the TOPIx procedure is followed for the removal and replacement of the original vehicle battery
8. Install the original vehicle battery within 72 hours of vehicle handover to the customer
9. Make sure if the display vehicle is being returned to the vehicle storage compound the original battery is installed (in a fully charged state), refit the transit relay (if required) and put the vehicle into transportation mode
10. Remove the slave battery identification tag JLR-415-010 from the vehicle keys once the original battery has been refitted
11. The original battery should be stored separately and clearly identified with the VIN from the vehicle from which it was removed.

TRANSIT RELAY INSTALLATION SUMMARY

	JAGUAR	LAND ROVER
Vehicles with Transit Relay installed	X152, X250 and X351	L319, up to 16 MY L405 and L494
Vehicles without Transit Relay installed	X760, X260 and X761	L316, L359, L538, L550, from 17 MY L405 and L494

2.2 SOFTWARE REFLASH, SDD WORK OR IGNITION ON RELATED WORKSHOP ACTIVITIES

Due to the high electrical current demand and high depth of discharge that can occur during vehicle software re-flash activities, SDD work or ignition on (power mode 6) related work in the workshop, vehicles that are undergoing such activities MUST have a JLR approved power supply capable of delivering 50 Amps or more. Approved showroom conditioners can be found on the JLR tooling website.

CAUTION:

> Do Not carry out any Software downloads with either or both the transit mode on or the transit relay fitted, this could damage vehicle modules.

2.3 EXTENDED VEHICLE REWORK

For any extended vehicle rework that results in consuming vehicle battery power a JLR approved power supply should be connected, refer to : Battery Support Unit connection procedure (414-00 battery and charging system - General information, General procedures).

2.4 JUMP STARTING NEW VEHICLES BEFORE THEY HAVE BEEN DELIVERED TO THE CUSTOMER

- It is the retailers responsibility to make sure the battery is not allowed to discharge by following the instructions and processes defined in this manual.
- However, if circumstances dictate that a new vehicle must be jump started due to a discharged battery whilst the vehicle is in the retailers care, **the battery on this vehicle must be replaced with a new one** prior to

delivery to the customer at the retailers liability.

- The vehicle should also undergo investigation as to why the battery became discharged.
- Do not connect the jump starting cable to the negative (-) terminal of the battery. Always connect to the recommended earth point. As defined in the owners handbook or service documentation for that vehicle.

2.5 AGM BATTERIES

- **AGM batteries must not be charged above 14.8 Volts. Doing so will damage them.**
- AGM Batteries must be tested with a capable battery tester as **detailed in the Equipment section (Section 5) of this procedure.**

NOTE:

Under no circumstances should the battery be disconnected with the engine running because under these conditions the generator can give a very high output voltage. This high transient voltage will damage the electronic components in the vehicle. Loose or incomplete battery connections may also cause high transient voltage.

3. HEALTH AND SAFETY PRECAUTIONS

WARNINGS:

- BATTERY CELLS CONTAIN SULPHURIC ACID AND EXPLOSIVE MIXTURES OF HYDROGEN AND OXYGEN GASES. IT IS THEREFORE ESSENTIAL THAT THE FOLLOWING SAFETY PRECAUTIONS ARE OBSERVED.
- Batteries emit highly explosive hydrogen at all times, particularly during charging. To prevent any potential form of ignition occurring

when working in the vicinity of a battery:
  - Do not smoke when working near batteries.
  - Avoid sparks, short circuits or other sources of ignition in the battery vicinity.
  - Switch off current before making or breaking electrical connections.
  - Ensure battery charging area is well ventilated.
  - Ensure the charger is switched off when: a) connecting to a battery; b) disconnecting from the battery.
  - Always disconnect the ground cable from the battery terminal first and reconnect it last.
- Batteries contain poisonous and highly corrosive acid. To prevent personal injury, or damage to clothing or the vehicle, the following working practices should be followed when topping up, checking electrolyte specific gravity, removal, refitting or carrying batteries:
  - Always wear suitable protective clothing (an apron or similar), safety glasses, a face mask and suitable gloves.
  - If acid is spilled or splashed onto clothing or the body, it must be neutralized immediately and then rinsed with clean water. A solution of baking soda or household ammonia and water may be used as a neutralizer.
  - In the event of contact with the skin, drench the affected area with water. In the case of contact with the eyes, bathe the affected area with cool clean water for approximately 15 minutes and seek urgent medical attention.
  - If battery acid is spilled or splashed on any surface of a vehicle, it should be neutralized and rinsed with clean water.
  - Heat is generated when acid is mixed with water. If it becomes necessary to prepare electrolyte of a desired specific gravity, SLOWLY pour the concentrated acid into water (not water into acid), adding small amounts of acid while stirring. Allow the

electrolyte to cool if noticeable heat develops. With the exception of lead or lead-lined containers, always use non-metallic receptacles or funnels. Do not store acid in excessively warm locations or in direct sunlight.

- Due to their hazardous contents, the disposal of batteries is strictly controlled. When a battery is scrapped, ensure it is disposed of safely, complying with local environmental regulations. If in doubt, contact your local authority for advice on disposal facilities.

4. BATTERY CARE REQUIREMENTS

4.1 RECEIPT OF A NEW VEHICLE

Within 24 hours of receipt of a new vehicle, a battery condition check must be carried out in accordance with the battery test process utilizing a JLR approved tester as outlined in **the Equipment section (Section 5) of this procedure**.

NOTES:

- The Midtronics software can be updated; refer to Administration bulletins to ensure the EXP-1080, GRX-3080 and GR1-3080 are up-to-date.
- Midtronics WiFi dongles are mandated; refer to special tool release notes titled “Wi-Fi pod upgrade to the EXP-1080 battery tester”. Ensure dongles are fitted at all times for the system to work correctly.
- The Midtronics code from the tester must be recorded on the form.

Any actions must be carried out in accordance with the table shown in the **Determining Battery Condition section (Section 6) of this procedure**. The

details must be recorded on the New Vehicle Storage Form which is part of the new vehicle storage document.
For additional information, refer to: New Vehicle Storage Form (100-11 Vehicle Transportation Aids and Vehicle Storage, Description and Operation).

4.2 NEW VEHICLE STORAGE

If the vehicle is to be stored, transport mode MUST be programmed into the vehicle with the transit relay fitted as delivered. Certain vehicles do not have a transit relay fitted, these vehicles only need transport mode installed, For additional information, refer to the **Transit Relay Installation Summary table** for transit relay installation.

Transit relay removal / vehicle placed in normal mode should only be completed a maximum of 72 hours prior to handover to customer

For vehicles without either a transit mode or transit relay the battery negative cable must be DISCONNECTED from the battery.

The battery must be tested and/or re-charged every 30 days and MUST be re-charged after every 90 day period.

NOTES:

- Transport mode is required to be set when a transit relay is installed to a vehicle to prevent damage to vehicle electrical modules.
- The Midtronics code from the tester must be recorded on the form.

Any actions must be carried out in accordance with the table shown in the **Determining Battery Condition section (Section 6) of this procedure**. The details must be recorded on the New Vehicle Storage Form which is part of the new vehicle storage document.
For additional information, refer to: New Vehicle Storage Form (100-11

Vehicle Transportation Aids and Vehicle Storage, Description and Operation).

4.3 PDI / DELIVERY TO CUSTOMER

Before the vehicle is handed over to the customer and as part of the PDI, the condition of the battery needs to be confirmed. The battery condition must be checked in accordance with the battery test process utilizing a JLR approved tester as outlined in **the Equipment section (Section 5) of this procedure**.

NOTE:

> The Midtronics code from the tester must be recorded on the form.

Any actions must be carried out in accordance with the table shown in the **Determining Battery Condition section (Section 6) of this procedure**. The details must be recorded on the New Vehicle Storage Form which is part of the new vehicle storage document.
For additional information, refer to: New Vehicle Storage Form (100-11 Vehicle Transportation Aids and Vehicle Storage, Description and Operation).

4.4 REPLACEMENT BATTERIES FOR SERVICE

All service replacement batteries must have the battery condition checked within 24 hours of receipt and controlled on a 'First In First Out' basis to ensure batteries are not allowed to age unnecessarily.

For batteries in storage and not yet fitted to a vehicle, they must be stored in a cool/dry environment, not in direct sunlight or under any direct heat source. Any batteries exhibiting any forms of damage or corrosion must not be fitted to any vehicle. Any batteries which are dropped must be scrapped, this applies even if no external damage is apparent.

The battery condition must be checked every 30 days in accordance with the battery test process utilizing a JLR approved tester as outlined in **the Equipment section (Section 5) of this procedure**.

Any actions must be carried out in accordance with the table shown in the **Determining Battery Condition section (Section 6) of this procedure**.The details must be recorded on the New Vehicle Storage Form which is part of the new vehicle storage document.
For additional information, refer to: New Vehicle Storage Form (100-11 Vehicle Transportation Aids and Vehicle Storage, Description and Operation).

4.5 BATTERY MAINTENANCE

- Any battery whether it is in a vehicle or a replacement part must be tested and/or re-charged every 30 days and MUST be re-charged after every 90 day period.
- The battery must always be connected to a battery support unit during any diagnostic sessions, including software updates.
  For additional information, refer to: Battery Support Unit Connection Procedure (414-00 Battery and Charging System - General Information, General Procedures).

4.6 BATTERY TEST

Enhancement to the recommended battery test equipment software, has removed the need to physically remove battery surface charge before completing a Battery Test.

The battery may be tested either on a bench or on the vehicle.

NOTE:

> Make sure the Ignition is OFF and the vehicle is powered down with the modules entering 'sleep mode' before commencing with the battery test.

Battery Test Types

Flooded Battery Testing

In some markets serviceable flooded Batteries are still in use, to test these correctly check the electrolyte level as outlined in the CONFIRMING ELECTROLYTE LEVEL section 9 in this procedure before any further testing. These batteries are identified with cell plugs on top of the battery.

All AGM and Flooded Battery Testing

The recommended Battery Tester has three types of Battery Test available for the technician to select:

1. Battery Test - The BATTERY TEST should be used on any battery that has started its warranty life cycle
   - The battery is in use and fitted to a vehicle registered to an owner
2. PDI / Storage - The PDI / STORAGE test should be used on any battery that has not yet been entered into the warranty life cycle
   - The battery is fitted to a **NEW** vehicle, but the vehicle has not yet been sold/registered to an owner
3. Battery Storage - The BATT. STORAGE test should be used on any battery that has not yet been entered into the warranty life cycle
   - The battery is not in use and is a parts stock battery and has not yet been fitted to a vehicle

NOTE:

> When BATT. STORAGE is selected, the technician must enter IDENTIFICATION data or a purchase order reference number for the battery being tested.

The battery condition must be checked in accordance with the battery test process utilizing a JLR approved tester as outlined in **the Equipment section (Section 5) of this procedure**.

NOTE:

> The Midtronics code from the tester must be recorded on the form.

Any actions must be carried out in accordance with the table shown in the **Determining Battery Condition section (Section 6) of this procedure**. The details must be recorded on the New Vehicle Storage Form which is part of the new vehicle storage document.
For additional information, refer to: New Vehicle Storage Form (100-11 Vehicle Transportation Aids and Vehicle Storage, Description and Operation).

CAUTION:

> DO NOT connect the tester to any other circuit or chassis point other than the battery negative terminal.

5. EQUIPMENT

All equipment used must be functionally capable of meeting the compliance requirements. Please refer to the approved equipment document (JLR 000015).

In the case of batteries fitted to a new vehicle at the retailer, battery condition should be measured using the appropriate hand-held Midtronics tester as follows:

BATTERY TYPE	BATTERY TESTER	BATTERY TESTER
	Jaguar	**Land Rover**
AGM & Flooded	Midtronics EXP1080, GRX-3080 & GR8-1180	Midtronics EXP1080, GRX-3080 & GR8-1180

The test results must be recorded on the New Vehicle Storage Form which is part of the new vehicle storage document.
For additional information, refer to: New Vehicle Storage Form (100-11 Vehicle Transportation Aids and Vehicle Storage, Description and Operation).

NOTE:

All equipment must be calibrated

6 DETERMINING BATTERY CONDITION

TESTER RESULTS	ACTION
GOOD BATTERY	Return to service.
CHARGE AND RE-TEST	Fully charge battery as per the time to charge table on the EPX-1080. Re-test battery. If same result replace battery.
REPLACE BATTERY OR BAD CELL BATTERY	Verify surface charge removed. Disconnect battery from vehicle and re-test. If result repeats after surface charge removal, replace battery. **DO NOT RECHARGE**.
UNABLE TO DO TEST	Disconnect battery from vehicle and re-test.

It is essential that a suitably ventilated defined area exists in each retailer for battery charging.

CAUTION:

It is very important that when charging batteries using the traction charger or other stand-alone chargers that the charger is set for the correct type of battery before charging commences. If the wrong switch is selected the result would be a battery that is not charged fully and / or overheating can occur. Follow the manufacturers operating instructions.

Batteries **MUST BE** tested and if necessary charged every 30 days and charged after 90 days irrespective of any test. It is recommended that retailers always have fully charged batteries ready for use.

CAUTION:

Do not charge AGM batteries with voltages over 14.8 Volts as this will damage the battery.

A designated controlled area must be allocated for scrap batteries and clearly controlled as such.

To bring a discharged but serviceable battery back to a fully charged condition proceed as follows:

- Check and if necessary top-up the battery electrolyte level. (Flooded maintainable batteries only)
- Charge the battery using a JLR approved charger as detailed in the approved equipment document following the manufacturers operating

instructions.

NOTE:

> When using the Midtronics Diagnostic Charger, automatic mode must always be used. After charging and analysis, the charger may display 'Top-Off Charging', Hit STOP To End. Do not stop charging until the current falls to 5A or less, otherwise the battery will not be fully charged.

Following charging, a post charge battery condition test must be carried out in accordance with the table shown in the **Determining Battery Condition section (Section 6) of this procedure**.

NOTE:

> The Midtronics code from the tester must be recorded on the form.

Any actions must be carried out in accordance with the table shown in the **Determining Battery Condition section (Section 6) of this procedure**. The details must be recorded on the New Vehicle Storage Form which is part of the new vehicle storage document.
For additional information, refer to: New Vehicle Storage Form (100-11 Vehicle Transportation Aids and Vehicle Storage, Description and Operation).

CAUTION:

AGM and Flooded batteries are NOT interchangeable, only specified batteries must be must. Failure to follow this instruction will result in early battery failure.

If it is determined that a battery requires replacement, always refer to the appropriate section of the workshop manual for instructions on removing and installing the battery from the vehicle.

On in service vehicles fitted with a Battery Monitoring System (BMS), the BMS control module must be reset following the installation of a new battery. The BMS control module reset procedure must be performed using an approved diagnostic system.

9 CONFIRMING ELECTROLYTE LEVEL

WARNINGS:

- BEFORE CHECKING THE ELECTROLYTE LEVEL AND TOPPING-UP THE BATTERY WITH DISTILLED WATER, REFER TO THE HEALTH AND SAFETY PRECAUTIONS SECTION.
- AGM TECHNOLOGY BATTERIES ARE FULLY SEALED FOR LIFE AND NO ATTEMPT SHOULD BE MADE TO CHECK OR TOP UP THE ELECTROLYTE LEVEL.

NOTES:

- In hot climates more frequent checks of the battery electrolyte level and condition are required. If necessary, the battery cells can be topped up using distilled water.
- Only flooded batteries can have the electrolyte level topped up.

On certain flooded battery types the electrolyte level will need to be checked.

1. Make sure the battery is of a type suitable for topping up. These types of batteries will have cell plugs visible on the top face of the battery or a removable access panel to allow access to the cells.
2. On batteries with a clear or opaque case and level marks, check the electrolyte level by visual inspection of the maximum level indicator mark on the battery casing, indicating adequate level above the battery separators.
3. On batteries with black cases, remove the cell plugs or access panel and ensure the electrolyte level is level with the indicator in the cell hole. A flashlight may be required to see the electrolyte level on this type of battery.
4. If the electrolyte level is low, top-up using **distilled water only**.
5. Refit the battery cell plugs.
6. Charge the battery for 1 hour using a recommended charger, if a GRX-3080 is used select manual charging for an hour.
7. Carry out a battery test using the recommended test equipment in line with **4.6 Battery Test & 5 Equipment**.

NOTE:

Maintenance free and Valve Regulated (AGM) batteries are sealed and therefore cannot be topped up.

CAUTION:

DO NOT overfill.